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exposition of the fundamentals of continuum mechanics, much of the book is concerned with classical theories and the mathematical techniques used in those theories.

The longer chapters are on viscous flow, incompressible and compressible inviscid flow, and infinitesimal elasticity. In addition there are chapters on viscoelasticity, plasticity, finite elasticity, and Reiner-Rivlin fluids, but there is quite properly less emphasis on these subjects.

The text is full of worked examples, which illustrate the use of a wide variety of mathematical techniques. There are many interesting problems at the end of each chapter, and a solution section in which these problems are worked out in detail. Because of this emphasis on problem-solving, the book gives a much more complete picture of the nature of continuum mechanics than those which cover the fundamentals alone.

Flow-Induced Vibration. By Robert D. Blevins. Van Nostrand, Reinhold, N. Y. 1977. Pages xiii-363. Cost $16.95.

REVIEWED BY R. H. SCANLAN[5]

This book represents the accumulated engineering experience of its author in dealing, mainly in an analytical manner, with a variety of vibration problems of structures under fluid-induced excitation or other fluid influence. After a brief introduction and a cursory look at dimensional analysis, the text deals in some depth with vortex-induced vibration, galloping vibrations and stall flutter, instabilities of tube rows and arrays, vibrations induced by oscillating flow and by turbulence, the damping of structures, sound induced by vortex shedding, vibrations of pipes containing fluid flow, and ship motion in a seaway. Short Appendices deal with modal analysis, principal coordinates, and aerodynamic sources of sound. Finally, a few unsolved problems are briefly listed.

The mechanical structures are treated, for the most part, as a linearly elastic. The chief aim, in each topic, is to derive for each phenomenon an analytical description or model the behavior of which can be brought into agreement with physical observations through recourse to experiment. The engineering usefulness of such calibrated models is then to predict the expected performance of actual structures. The author makes reference to modern literature dealing with the topics treated.

While in the opinion of the reviewer the book constitutes a conscientious, useful, and very timely review of methods in the field, it will not necessarily respond to the whole gamut of needs, nor please every reader. It nonetheless can be of strong help to the engineer confronted with like problems in present-day technology.

Treatments of the various topics are good, but necessarily reflect the biases of the author and are therefore selective rather than exhaustive. Dealing with vortex shedding, for example, while effective descriptive coverage is provided, the author bypasses the well-known Hartlen-Currie lift-oscillator model (and other models) in favor of one of his own (with Iwan). Some appropriate attention is, however, awarded to treatment of the three-dimensional vortex shedding phenomenon as random, though again without full reference to other treatments (such as that of Vickery and Clark, 1972) than the author's own.

Under "Galloping Vibrations and Stall Flutter," the author recreates the models of Den Hartog, Parkinson, and Novak and also points out the role of torsion and general two-degree-of-freedom action in this instability phenomenon. Again, the coverage is generally good, though some topics are slighted or mistreated.

In the latter category is bridge flutter, not, admittedly, a primary concern of the author. Neither does his experience extend to classical flutter, which facts may account for two casually erroneous remarks in the text. The first (p. 10) is "These vibrations [galloping] are similar to classical aircraft flutter except that they are self-limiting," and the second (p. 78) is "One is tempted to apply airfoil theory to relatively smooth sections, in order to analytically predict the steady forces on the section and avoid the necessity for static testing (reference 4-25)."

In the first instance it is abundantly clear in the literature that galloping is a phenomenon well modeled on the basis of (nonlinear) statically measured coefficients, while classical flutter is, on the other hand, modeled well only by motion-dependent coefficients. In the second instance the reference given is casually misjudged since that reference deals in fact with information (indicial lift functions of bridges) specifically derived from the measured oscillatory derivatives of those structures (which are not smooth), and not from airfoil theory, nor from static force coefficients.

The topic of instabilities of tube rows and arrays is currently of increased importance in view of the stepped-up performance demands on such devices as modern heat exchangers, notably for high-output power plants. Some models are given, and the chapter serves as a good introduction for the engineer entering the field. Wake-induced vibration is also discussed, following the models of Simpson, and of Wardlaw, Cooper, and Scanlan. Chapter references do not include many possible others in the wide literature concerned with tube and wake-induced instability.

Under "Vibrations Induced by Oscillating Flow," the author deals with in-line and wave forces. Treatment is good, but brief, and the engineer seeking, for example, a broad treatment of wave forces on offshore structures will have to look elsewhere. However, the approaches used are solid and introduce a number of the parameters of importance.

The chapter on "Vibrations Induced by Turbulence" employs the elements of random vibration theory, and makes much use of the generalized type of approach first employed by Eringen (1957) though this reference is not given. The chapter briefly treats a number of topics such as the response of a flexible rod in a turbulent flow, buffeting of tube arrays, and vibrations produced by winds. The latter topic provides the reader with only a glimpse of the vast modern wind engineering literature, but it may serve as one kind of introduction to that literature, most of which, of necessity, is not cited.

Damping, both from fluid sources and solid sources, is discussed in a separate chapter. Much more could be said on the subject, and the treatment, again, is far from exhaustive. Some useful numerical values of damping ratio have been collected and included from a variety of sources. Bridge damping is mentioned without allusion to the strong modifying effect of the surrounding air.

Chapter 9 provides an introduction to sound produced by vortex shedding. Chapter 10 treats vibrations of a pipe with axial flow, a topic of strong interest in modern mechanical engineering. Chapter 11, somewhat of a departure from the other subject matter treated, deals in fair detail with ship motion in waves, and a brief list of references is given.

Overall, the book may be seen as a very timely account of the author's experience in dealing analytically with a group of problem-types that are, for the most part, naturally linked together. The treatments are generally responsible, providing good introductions to modern viewpoints for the serious (and perhaps preferably, advanced) engineer. The aims throughout are practical. The book is clearly intended as introductory (though not necessarily elementary), not as an exhaustive treatise. In this light it generally fulfills its mission. The author has provided a useful addition to the reference shelf of the working engineer-analyst confronted with a problem that falls in the areas treated.

The book helps bring the profession a step nearer to the day when, in light of the considerable analytic effort expended on the (mostly unwanted) phenomena of flow-induced vibration, the engineer may look ahead, directly from the earliest stages of design, toward design strategies that are knowledgeably aimed at avoiding these phenomena.

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